

NESTED QUANTIFIERS

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Nested Quantifier:

one quantifier is within the scope of another quantifier
such as

$$\forall x \forall y P(x, y)$$

$$\forall x \exists y P(x, y)$$

$$\exists x \forall y P(x, y)$$

$$\exists x \exists y P(x, y)$$

Nested Quantifiers commonly occur in mathematics and Computer Science.

- We will see how to use nested Quantifiers to express mathematical stmts.
- We will show how nested quantifiers can be used to translate english sentences.

UNDERSTANDING STATEMENTS INVOLVING QUANTIFIERS (Quantified Logical Expression into English Sentences)

Example # 1(a): (CASE 1: $\forall x \forall y$)

Consider the following quantified logical expression:

$$\forall x \forall y (x + y = y + x) \quad \text{where domain for variables } x \text{ \& } y \text{ consists of all real numbers.}$$

Translate the above statement into english sentence.
And what is the truth value of this statement.

Sol:

ol:
Let

domain : real numbers

$$\forall x \forall y (x+y = y+x)$$

The above statement is true for all real numbers x and y .

i.e., Commutative Law for addition of real numbers

$i=1$ $j=1$ to n	$i=2$ $j=1$ to n		$i=n$ $j=1$ to n
$P(x_1, y_1) : T$	$P(x_2, y_1) : T$	...	$P(x_n, y_1) : T$
$P(x_1, y_2) : T$	$P(x_2, y_2) : T$	...	$P(x_n, y_2) : T$
$\equiv$	$\equiv$		$\equiv$
$P(x_1, y_n) : T$	$P(x_2, y_n) : T$	...	$P(x_n, y_n) : T$

$x+y = y+x$ is true for all x and y .

So

$$\forall x \forall y (x+y = y+x) \text{ is true}$$

The order of Quantification

If we change the order of quantification, the meaning of the statement will be the same. i.e.,

$\forall x \forall y (x+y = y+x) \neq \forall y \forall x (x+y = y+x)$
are equivalent.

Conclusion :

The order of nested Universal Quantifiers in a statement can be changed without changing the meaning of the statement.

(Order is not important if both are Universal quantifiers)

CONCEPT

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$$\forall x \forall y P(x, y)$$

$P(x, y)$ is true for all real number x and y

THINKING QUANTIFICATION AS LOOPS :

$\forall x P(x) :$

```
for i=1 to n
  if ( $\neg P(x_i)$ )
    return false
return true
```

$\forall x P(x)$ is true

if $P(x)$ is true for every x in D (domain).

$\forall x P(x)$ is false

if $P(x)$ is false for at least one x in D .

$\forall x \forall y P(x, y) :$

```
for i=1 to n
  for j=1 to m
    if ( $\neg (P(x_i, y_j))$ )
      return false
return true
```

We loop through the values for x , and

for each x , we loop through the values for y .

If we ever hit a value x for which we hit a value y for which $P(x, y)$ is false, then

$\forall x \forall y P(x, y)$ is false.

So, $\forall x \forall y P(x, y)$ is true

if $P(x, y)$ is true for all values of x & y .

Example #1(b)

Consider the statement

$$\forall x \forall y \forall z (x + (y + z) = (x + y) + z) \quad \text{domain: all variables consist of real numbers}$$

Translate the above statement into english sentence.

Sol:

$$\forall x \forall y \forall z (x + (y + z) = (x + y) + z)$$

Here

$$x + (y + z) = (x + y) + z \text{ is true}$$

for all real numbers x, y and z

i.e.; Associative Law for addition of real numbers

```

for i=1 to n
  for j=1 to n
    for k=1 to n
      if (¬(P(xi, yj, zk)))
        return false
    return true
  
```

Example #2 : (CASE 2 : $\forall x \exists y$)

Consider the statement

$$\forall x \exists y (x + y = 0)$$

domain: all variables consist of real numbers

Translate the above quantified logical expression into English sentence. What is the truth value of this statement

Sol:

sl:

Let

$$\forall x \exists y (x+y=0) \quad \text{domain: real numbers}$$

As every real number has an additive inverse,

$$\text{i.e. } 5 + (-5) = 0$$

$$-7 + (7) = 0$$

$\equiv$
etc.

So, we can say that

For every real number x , there is a real number y for which $(x+y=0)$ is true.

So,

$$\forall x \exists y (x+y=0) \text{ is true}$$

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CONCEPT

$$\forall x \exists y P(x, y)$$

For every real number x , there is a real number y for which $P(x, y)$ is true.

THINKING AS LOOPS:

$\exists y P(y)$:

```
for i=1 to n
  if ( P(yi) )
    return true
return false
```

$P(y)$ is true,
if $P(y)$ is true for
at least one x in D
and false
if $P(y)$ is false for
every x in D

$$\forall x \exists y P(x, y):$$

```
for i=1 to n
  for j=1 to n
    if ( P(xi, yj) )
      return true
return false
```

We loop through the values for x , and for each x , we loop through the values for y until we find a y for which $P(x, y)$ is true.

If for every x , we hit such a y , then

$$\forall x \exists y P(x, y) \text{ is true}$$

If for some x , we never hit such a y , then

$$\forall x \exists y P(x, y) \text{ is false}$$

Example #3 : (CASE 3 : $\exists y \forall x$)

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Consider the Statement of example 2 with slight modifications as:
(changing the order)

$$\exists y \forall x (x+y=0) \quad \text{domain: real numbers}$$

What is the truth value of the above statement?

Sol :

$$\exists y \forall x (x+y=0)$$

means

There is a real number y such that for every real number x for which $(x+y=0)$ is true.

Here,

No matter what the value of y is chosen, there is only one value of x for which $(x+y=0)$ is true.

$$\begin{aligned} \dots + 4 &\neq 0 \quad F \\ -6 + 4 &\neq 0 \quad F \\ -5 + 4 &\neq 0 \quad F \\ -4 + 4 &= 0 \quad \text{true} \\ -3 + 4 &\neq 0 \quad F \end{aligned}$$

As there is no real number y such that $(x+y=0)$ is true for all real numbers x

So $\exists y \forall x (x+y=0)$ is false

Conclusion

From Case 2 and 3, we conclude that

if different quantification is there in a statement $(\exists \neq \forall)$, then the order is important i.e.,

$\forall x \exists y P(x,y)$ and $\exists y \forall x P(x,y)$ are not logically equivalent.

$$\exists y \forall x P(x,y)$$

There is a real number y such that for every real number x for which $P(x,y)$ is true.

We loop through the values for y until we find a y for which $P(x,y)$ is always true when we loop through all values for x .

once we find such a y ,

$$\exists y \forall x P(x,y) \text{ is true}$$

If we never hit such a y , then we know that

$$\exists y \exists x P(x,y) \text{ is false.}$$

$$\forall x \exists y P(x,y):$$

True: for every x , there is a y for which $P(x,y)$ is true

False: there is an x such that $P(x,y)$ is false for every y

$$\exists y \forall x P(x,y):$$

True: there is a y for which $P(x,y)$ true for every x .

False: For every x there is a y for which $P(x,y)$ false

OBSERVATIONS

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$$\forall x \exists y P(x, y):$$

$\forall x \exists y P(x, y)$ is true iff for every value of x there is a value of y for which $P(x, y)$ is true.

So, for this statement to be true, no matter which x you choose, there must be a value of y (possibly depending on the x you choose) for which $P(x, y)$ is true.

On the other hand

$$\exists y \forall x P(x, y):$$

$\exists y \forall x P(x, y)$ is true iff there is a y that makes $P(x, y)$ true for every x .

So, for this statement to be true, there must be a particular value of y for which $P(x, y)$ is true regardless of the choice of x .

From these observations,

It follows that

if $\exists y \forall x P(x, y)$ is true, then $\forall x \exists y P(x, y)$ must also be true.

However,

if $\forall x \exists y P(x, y)$ is true, it is not necessary for $\exists y \forall x P(x, y)$ to be true.

Example: Truth values of $\begin{cases} \forall x \forall y \exists z Q(x, y, z) \\ \exists z \forall x \forall y Q(x, y, z) \end{cases}$ $Q(x, y, z): x + y = z$
domain: all are real

Sol:-
• $\forall x \forall y \exists z (x + y = z)$ is true, As
① any ② any ③ There exist
 $x + y = z$

• $\exists z \forall x \forall y (x + y = z)$ is false

there is no value of z that satisfies the above for all values of x & y .
for all for all ① fixed
 $x + y = z$

(CASE 4 : $\exists x \exists y$)

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$$\exists x \exists y P(x, y)$$

There is a pair x, y for which $P(x, y)$ is true.

We loop through the values of x , where for each x we loop through the values for y until we hit an x for which we hit a y for which $P(x, y)$ is true.

The statement $\exists x \exists y P(x, y)$ is false only if we never hit an x for which we hit a y such that $P(x, y)$ is true.

Note:

As in the case of both universal quantification, the order is not important if both quantifiers are existentials.

i.e

$\exists x \exists y P(x, y)$ and $\exists y \exists x P(x, y)$ are equivalent.

If we change the order of quantification, the meaning will be the same.

$$\exists x \exists y P(x, y)$$

$$\exists y \exists x P(x, y)$$

TRUE: There is a pair x, y for which $P(x, y)$ is true.

FALSE: $P(x, y)$ is false for every ~~$P(x, y)$~~ x, y .

TRANSLATING FROM LOGICAL EXPRESSION INTO ENGLISH 8 of 16

- The first step is to write out what the quantifiers and predicates in the expression mean.
- The next step is to express this meaning in a simpler sentence.

Example #1 :

Translate the statement

$$\forall x (C(x) \vee \exists y (C(y) \wedge F(x, y)))$$

Where $C(x)$: x has a Computer (similarly $C(y)$: y has a Computer)

$F(x, y)$: x and y are friends

domain: x and y consists of all students in your school.

Sol: Let

$$\forall x (C(x) \vee \exists y (C(y) \wedge F(x, y)))$$

First Step: Write out what the quantifiers and predicates mean

" For every student x in your school, x has a Computer or there is a student y such that y has a Computer and x & y are friends.

NEXT STEP: Express in simpler sentence.

Translation { Every student in your school

- has a Computer
- has a friend who has Computer

Example 2 :

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Translate the statement

$$\exists x \forall y \forall z \left((F(x, y) \wedge F(x, z) \wedge (y \neq z)) \rightarrow \neg F(y, z) \right)$$

where $F(a, b)$: a and b are friends

domain : x, y, z consists of all students in your school

Sol :

$$\exists x \forall y \forall z \left((F(x, y) \wedge F(x, z) \wedge (y \neq z)) \rightarrow \neg F(y, z) \right)$$

There is a student x such that for all students y and all student z other than y , **if** x and y are friends and x and z are friends, **then** y and z are not friends.

Simpler English sentence :

" There is a student none of whose friends are also friends with each other. "

TRANSLATING STATEMENTS INTO LOGICAL EXPRESSIONS

A: Mathematical statements (in english) into Logical Expressions

B: English Sentences into Logical Expressions.

A: Mathematical (English) Statements into Logical Expression

Example #1:

Translate the mathematical statement (expressed in english) into Logical expression.

"The sum of two positive integers is always positive"

Sol:

To identify the quantifiers, we rewrite the statement as:

1. Consider the domain of positive integers

For every positive integers x and y , $x+y$ is positive
it can be expressed as

$$\forall x \forall y (x+y > 0) \quad , \text{ where domain: all positive integers}$$

2. Consider the domain of all integers.

For every two integers x and y , if x and y are both positive, then the sum of x and y is positive.

It can be expressed as:

$$\forall x \forall y ((x > 0) \wedge (y > 0) \rightarrow (x+y) > 0)$$

domain: x, y consists of all integers

Example #2:

Translate the mathematical statements (expressed in english) into Logical expression.

"Every real number except zero has multiplicative inverse."

Def. Multiplicative Inverse:

A multiplicative inverse of a real number x is a real number y such that $xy = 1$

Sol:

To identify the quantifiers, we rewrite the statement as:

"For every real number x except zero, x has multiplicative inverse."

or

"For every real number x , if $x \neq 0$, then there exists a real number y such that $xy = 1$ "

It can be expressed logically:

$$\forall x (x \neq 0 \rightarrow \exists y (xy = 1))$$

Note:

We can move $\exists y$ to the left so that it appears just after $\forall x$, because y does not appear in left side of Conditional.

final Ans:

$$\forall x \exists y (x \neq 0 \rightarrow (xy = 1))$$

B: TRANSLATING ENGLISH SENTENCES INTO LOGICAL EXPRESSION.

EXAMPLE #1:

Translate the following sentence into Logical Expression

"If a person is female and is a parent, then this person is Someone's mother."

with the domain of all people.



Sol:

"If a person is female and is a parent, then this person is Someone's mother."

rewrite as

{ For every person x , if person x is female and person x is a parent, then
-there exists a person y , such that person x is the mother of person y . }

Let

$F(x)$: x is female

$P(x)$: x is parent

$M(x, y)$: x is mother of y

domain: all people

$$\forall x \left((F(x) \wedge P(x)) \rightarrow \exists y M(x, y) \right)$$

$\exists y$ can be moved to the left (it appears just after $\forall x$,
 $\therefore y$ does not appear in $(F(x) \wedge P(x))$, independent of this expression), so

we can write it as:

$$\forall x \exists y \left((F(x) \wedge P(x)) \rightarrow M(x, y) \right)$$

Example #2

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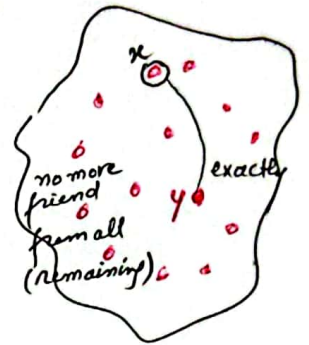
Translate the following into logical expressions:

"Everyone has exactly one best friend."

domain: all people

Sol:

"Everyone has exactly one best friend."



For every person x , there exist a person y who is the best friend of x , and
 (To maintain no other friend from all other than y of x)
 for every person z , if person z is not person y (~~already y is the best friend of x~~), then
 z is not the best friend of x .

$$\forall x \exists y (B(x, y) \wedge \forall z (z \neq y \rightarrow \neg B(x, z)))$$

~~where $B(a, y) : y$ is the best friend~~

$B(a, b) : b$ is the best friend of a

domain: all people.

"Everyone has exactly one best friend"

Example #3

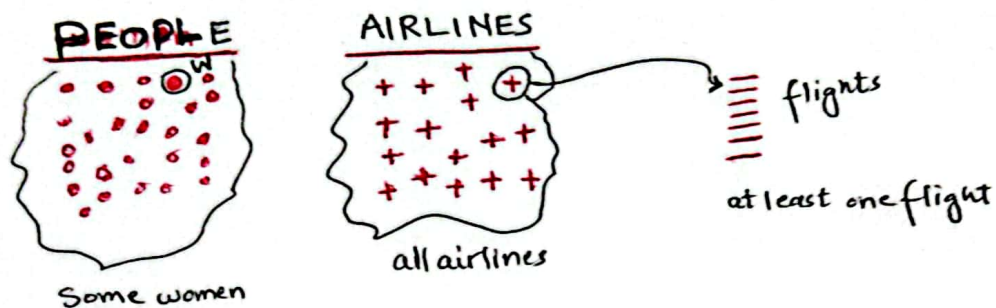
Translate the following into logical expression

"There is a woman who has taken a flight ~~a flight~~ on every airline in the world."

Sol:

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"There is a woman who has taken a flight on every airline in the world."



$P(w, f) : w$ has taken flight f

$Q(f, a) : f$ is a flight on a

$$\exists w \forall a \exists f (P(w, f) \wedge Q(f, a))$$

where domain :

w consists all women,

f all ~~airplane~~ flights,

a all airlines

Compact form:

$$\exists w \forall a \exists f, R(w, f, a)$$

where

$R(w, f, a) : w$ has taken f on a .

First Solution is usually preferable.

Second one somewhat obscures the relationships among the variables.

NEGATING NESTED QUANTIFIERS

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EXAMPLE #1 :

Express the negation of the statement

$$\forall x \exists y (xy = 1)$$

so that no negation precedes a quantifier.

Sol :

$$\forall x \exists y (xy = 1)$$

Apply the negation sign :

$$\neg \forall x \exists y (xy = 1) = \exists x \neg \exists y (xy = 1)$$

$$\text{As } \neg \forall x P(x) = \exists x \neg P(x)$$

$$= \exists x \forall y \neg (xy = 1)$$

$$= \exists x \forall y (xy \neq 1)$$

Example #2 :

Use quantifiers to express the statement that

"There does not exist a woman who has taken a flight on every airline in the world."

Sol :

By Example # , we have quantified statement of the following :

"There exists a woman who has taken a flight on every airline in the world."

The quantified statement : (we have found in Previous ~~ex~~ example)

$$\exists w \forall a \exists f (P(w, f) \wedge R(f, a))$$

Now Apply the $\neg$ to get the required result

$$\neg \exists w \forall a \exists f (P(w, f) \wedge Q(f, a))$$

$$\begin{aligned} \neg \exists w \forall a \exists f (P(w, f) \wedge Q(f, a)) &= \forall w \neg \forall a \exists f (P(w, f) \wedge Q(f, a)) \\ &= \forall w \exists a \neg \exists f (P(w, f) \wedge Q(f, a)) \\ &= \forall w \exists a \forall f \neg (P(w, f) \wedge Q(f, a)) \\ &= \forall w \exists a \forall f (\neg P(w, f) \vee \neg Q(f, a)) \end{aligned}$$

i.e.,

"For Every woman there is an airline such that for all flights, this woman has not taken that flight or that flight is not on this airline."